Assignment 6.

1. (a) $6x^4 + y^5 = 2x$

$\frac{d}{dx}: 24x^3 + 5y^4 y' = 2 \Rightarrow y' = \frac{2 - 24x^3}{5y^4}$

(b) $x^2 - 4xy + y^2 = 4$

$2x - 4y - 4xy' + 2yy' = 0 \Rightarrow y' = \frac{2y - x}{y - 2x}$

(c) $x^4 + x^2y^2 + y^3 = 5$

$4x^3 + 2xy^2 + 2x^2yy' + 3y^2y' = 0 \Rightarrow y' = -\frac{2x(2x^2+y^2)}{y(2x^2+3y)}$

(d) $\frac{x^2}{x+y} = y^2 + 1$

$\frac{2x(x+y) - x^2(1+y')}{(x+y)^2} = 2yy' \Rightarrow y' = \frac{x(x+2y)}{2x^2y + 4xy^2 + 2y^3 + x^2}$

(e) $xe^y = x - y$

$e^y + xe^y y' = 1 - y' \Rightarrow y' = \frac{1 - e^y}{xe^y + 1}$

(f) $\sin x + \cos y = 2x - 3y$

$\cos x - \sin y \cdot y' = 2 - 3y' \Rightarrow y' = \frac{2 - \cos x}{3 - \sin y}$

(g) $\sin(xy) = \cos(x+y)$

$\cos(xy) \cdot (y + xy') = -\sin(x+y)(1+y')$

$\Rightarrow y' = -\frac{y\cos(xy) + \sin(x+y)}{x\cos(xy) + \sin(x+y)}$

(h) $xy = \sqrt{x^2+y^2}$

$y + xy' = \frac{1}{2}(x^2+y^2)^{-\frac{1}{2}} \cdot (2x + 2yy')$

$\Rightarrow y' = \frac{x - y\sqrt{x^2+y^2}}{x\sqrt{x^2+y^2} - y}$

2. (a) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$

$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}y' = 0 \Rightarrow y' = -\frac{\sqrt[3]{y}}{\sqrt[3]{x}}$

When $x = -3\sqrt{3}$, $y = 1$, we have $y' = \frac{1}{\sqrt{3}}$.

So the tangent line is $y = \frac{1}{\sqrt{3}}(x + 3\sqrt{3}) + 1 = \frac{x}{\sqrt{3}} + 4$.

(b) $x^2 - xy - y^2 = 1$

$2x - y - xy' - 2yy' = 0 \Rightarrow y' = \frac{2x - y}{x + 2y}$

substitute $(2, 1)$, obtain $y' = \frac{3}{4}$

So the tangent line is $y = \frac{3}{4}(x - 2) + 1 = \frac{3}{4} - \frac{1}{2}$

(c) $x^2 + y^2 = (2x^2 + 2y^2 - x)^2$

$2x + 2yy' = 2(2x^2 + 2y^2 - x) \cdot (4x + 4yy' - 1)$

$\Rightarrow$ when $x = 0$, $y = \frac{1}{2}$, we have $y' = 1$

The tangent line is $y = x + \frac{1}{2}$.

3. $2(x^2 + y^2)^2 = 25(x^2 - y^2)$ —— eqn①.

$4(x^2 + y^2)(2x + 2yy') = 25(2x - 2yy')$

$\Rightarrow y' = \frac{25x - 4x(x^2 + y^2)}{25y + 4y(x^2 + y^2)}$

When the tangent line is horizontal, $y' = 0$

$\Rightarrow 25x - 4x(x^2 + y^2) = 0 \Rightarrow x^2 + y^2 = \frac{25}{4}$ —— eqn ②

(Note: When $x = 0$, $y = 0$, $y'(0) = \lim_{h \to 0} \frac{\Delta y}{h}$.

Substitute $\Delta y, h$ to the lemniscate, $y'(0) = \pm 1$, not horizontal)

Solve eqn ① and eqn ②, we have $x^2 = \frac{75}{16}$ and $y^2 = \frac{25}{16}$.

So the four points are $(\pm \frac{5\sqrt{3}}{4}, \pm \frac{5}{4})$

4. $\sqrt{x}+\sqrt{y}=\sqrt{c}$

$\frac{1}{2\sqrt{x}}+\frac{y'}{2\sqrt{y}}=0 \Rightarrow y'=-\frac{\sqrt{y}}{\sqrt{x}}$

An equation of a tangent line at (x_0, y_0) is

$y=-\frac{\sqrt{y_0}}{\sqrt{x_0}}(x-x_0)+y_0$.

So the x-intercept is $(x_0+\sqrt{x_0}\sqrt{y_0}, 0)$, and

y-intercept is $(0, y_0+\sqrt{x_0}\sqrt{y_0})$

The sum is $x_0+2\sqrt{x_0y_0}+y_0=(\sqrt{x_0}+\sqrt{y_0})^2=(\sqrt{c})^2=c$.

5. $f(x)=\frac{x-2}{x^2+1}$ on $[-1,2]$

$f'(x)=\frac{1\cdot(x^2+1)-(x-2)(2x)}{(x^2+1)^2}=\frac{-x^2+4x+1}{(x^2+1)^2}$

Find the critical points, $x=2\pm\sqrt{5}$.

Because $2+\sqrt{5}>2$, we only consider $2-\sqrt{5}$.

$f(-1)=-\frac{3}{2}$

$f(2-\sqrt{5})=-\frac{\sqrt{5}}{8-4\sqrt{5}}=\frac{5+\sqrt{5}}{2}$

$f(2)=0$

So the absolute maximum is $\frac{5+\sqrt{5}}{2}$ at $x=2-\sqrt{5}$.

also a local maximum.

The absolute minimum is $-\frac{3}{2}$ at $x=-1$.

There is no local minimum on this interval.

6. $f(x) = x^3 + bx^2 + cx + d$.

$f'(x) = 3x^2 + 2bx + c = 0$.

The critical points are given by the solutions. $x_{1,2} = \frac{-2b \pm \sqrt{4b^2 - 12c}}{6}$.

The discriminant is $4b^2 - 12c \geq 0 \Rightarrow c \leq \frac{b^2}{3}$.

So. if $c<0$, the function $f(x)$ can have local extrema.

$f''(x) = 6x + 2b$, if $c<0$, then $f''(x_{1,2}) = \pm\sqrt{4b^2 - 12c}$

one is positive and one is negtive. Therefore, $f(x)$ has both a local maximum and a local minimum.

7. $f(x) = x^4 + cx^2 + 1$

$f'(x) = 4x^3 + 2cx = 0 \Rightarrow x = 0$, $x = \pm\sqrt{-\frac{c}{2}}$ (if $c<0$)

$f''(x) = 12x^2 + 2c$

So, if $c>0$, there is a local minimum at $x=0$, $(f''(0)>0)$

if $c=0$, $f'(x)>0$ when $x>0$, $f'(x)<0$ when $x<0$.

$x=0$. is a local minimum point.

if $c<0$, $f''(0)<0$, $f''(\pm\sqrt{\frac{-c}{2}}) = -4c > 0$.

there is a local maximum at $x=0$.

two minimums at $x = \pm\sqrt{\frac{-c}{2}}$.

8. $y = \arcsin(1 - \frac{1}{x^2})$

The domain is determined by $-1 \leq 1 - \frac{1}{x^2} \leq 1 \Rightarrow x \geq \frac{1}{\sqrt{2}}$ or $x \leq -\frac{1}{\sqrt{2}}$.

$y' = \frac{1}{\sqrt{1 - (1 - \frac{1}{x^2})^2}} \cdot 2x^{-3} = \frac{2}{x\sqrt{2x^2+1}}$

There is no critical numbers.

Bonus.

1. (a) Let $f(x) = e^x - (1+x)$, we know $f(0) = 0$

$f'(x) = e^x - 1$, If $x > 0$, $f'(x) > 0$

$f(x)$ is increasing if $x > 0$.

So we can conclude that $f(x) > f(0)$ for $x > 0$.

In other words, $e^x - (1+x) > 0 \Rightarrow e^x > 1 + x$.

(b). Let $g(x) = e^x - (1 + x + x^2/2)$, do the same thing.

$g(0) = 0$, $g'(x) = e^x - (1+x)$.

We know $g'(x) > 0$ from item (a), g is increasing.

So $g(x) > g(0)$ when $x > 0$, so $e^x > 1 + x + \frac{x^2}{2}$.

(c). We can use the mean value theorem for $f(x) = \sin x$.

$$f(x) - f(0) = f'(c)(x - 0),$$

$$\Rightarrow |\sin x| = |\cos c|\,|x| \le |x|$$

(d) According to the mean value theorem.

$$\min_{a \le x \le b} f'(x) \le \frac{f(b) - f(a)}{b - a} = f'(c) \le \max_{a \le x \le b} f'(x)$$

If $f(x) = \cos x$, then $f'(x) = -\sin x$,

$$\min_{a \le x \le b} -\sin x \le \frac{\cos b - \cos a}{b - a} \le \max_{a \le x \le b} -\sin x$$

We know $-1 \le -\sin x \le 1$, so

$$\left| \frac{\cos b - \cos a}{b - a} \right| \le 1$$

$$\Rightarrow |\cos b - \cos a| \le |b - a|$$